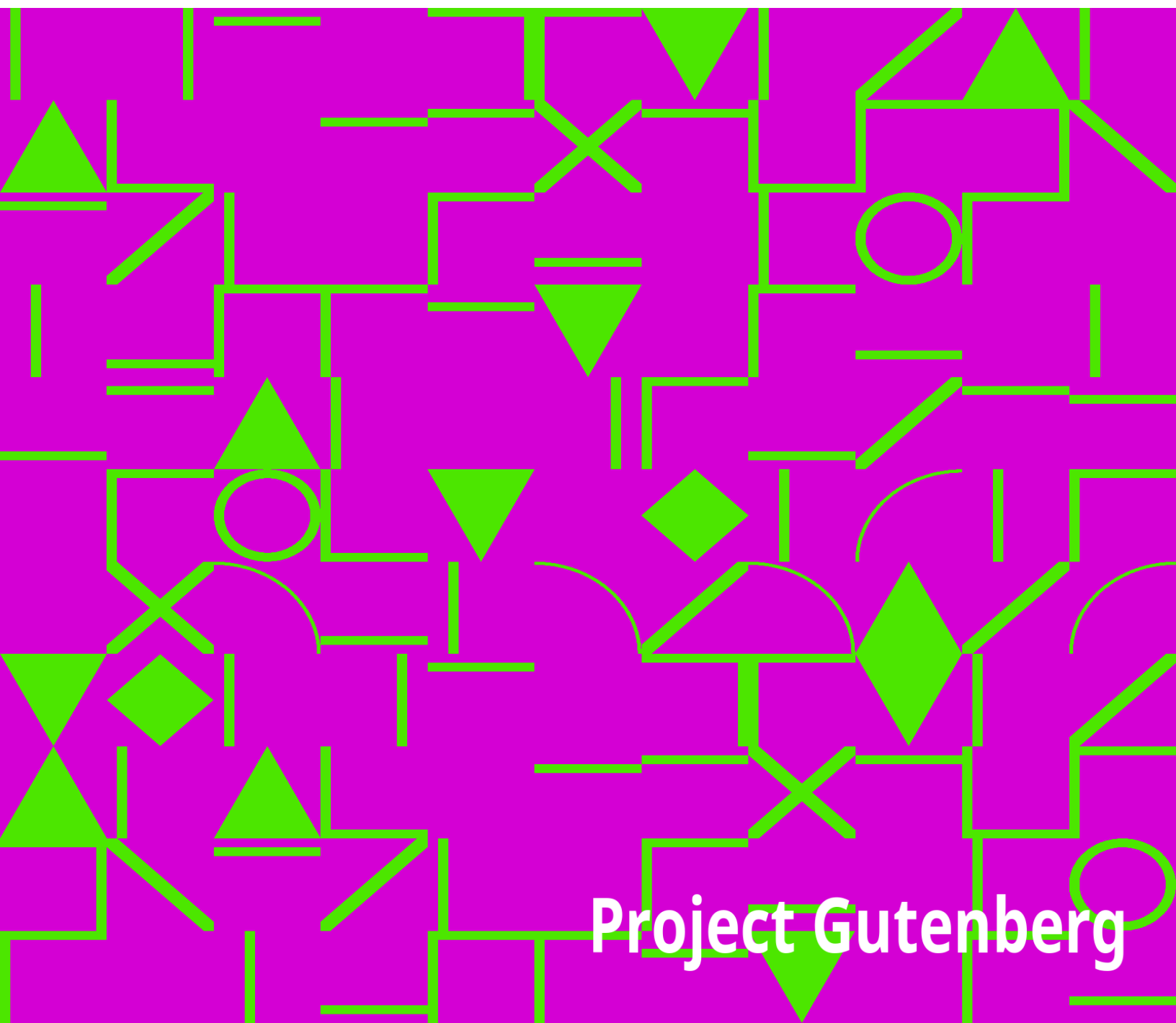


The Postnatal Development of Two Broods of Great Horned Owls (*Bubo virginianus*)

Donald Frederick Hoffmeister and Henry



Project Gutenberg

The Postnatal Development of Two Broods of Great Horned Owls (*Bubo virginianus*)

Donald Frederick Hoffmeister and Henry

An abstract geometric pattern consisting of various shapes like squares, triangles, circles, and lines, all in shades of blue and green, creating a complex, interlocking design.

Project Gutenberg

The Project Gutenberg eBook of The Postnatal Development of Two Broods of Great Horned Owls (*Bubo virginianus*)

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Postnatal Development of Two Broods of Great Horned Owls (*Bubo virginianus*)

Author: Donald Frederick Hoffmeister
Henry W. Setzer

Release date: January 31, 2011 [eBook #35118]

Language: English

Other information and formats: www.gutenberg.org/ebooks/35118

Credits: Produced by Chris Curnow, Joseph Cooper, Josephine Paolucci and the Online Distributed Proofreading Team at <http://www.pgdp.net>.

*** START OF THE PROJECT GUTENBERG EBOOK THE POSTNATAL
DEVELOPMENT OF TWO BROODS OF GREAT HORNED OWLS (*BUBO*
VIRGINIANUS) ***

THE POSTNATAL DEVELOPMENT OF TWO

BROODS OF GREAT HORNED OWLS

(*Bubo virginianus*)

BY

**DONALD F. HOFFMEISTER AND HENRY W.
SETZER**

University of Kansas Publications

Museum of Natural History

Volume 1, No. 8, pp. 157-173

October 6, 1947

UNIVERSITY OF KANSAS
LAWRENCE
1947

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, Donald S. Farner, H. H. Lane,
Edward H. Taylor

Volume 1, No. 8, pp. 157-173

October 6, 1947

UNIVERSITY OF KANSAS
Lawrence, Kansas

PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1947

21-6958

The Postnatal Development of Two Broods of Great Horned Owls

(*Bubo virginianus*)

By

DONALD F. HOFFMEISTER AND HENRY W. SETZER

Opportunity regularly to observe at the nest the development of young Great Horned Owls, *Bubo virginianus* (Gmelin), under favorable conditions, was afforded when a pair nested and reared their three offspring in 1945 and one offspring in 1946 on the vine-covered north wall of the Museum of Natural History at the University of Kansas. The observations here reported are based primarily on the three young raised in 1945 when daily observations were made. These have been supplemented by other observations made of the one nestling in 1946. Unless otherwise stated, observations pertain to the nest and three young in 1945.

NEST SITE

In 1945 the nest was situated on a metal-covered cement ledge, two feet wide and 48 feet above the ground, at the northeast corner of the Museum Building. The nest was protected on the east by a stone abutment of the building and on the south by the north wall of the building itself. Here the nest could be observed at will through a laboratory window without disturbing the birds. The taking of notes was begun at the time of egg-laying and extended to the time at which the young left the nest, February 3 through April 26, 1945. In 1946 the owls nested farther down the north side of the building, behind two cement pillars, approximately 25 feet above the ground. To examine the nest in 1946 it was necessary to lower an observer down the side of the building by means of a rope. Observations of this nest were never made more frequently than every other day. The adult owls were first seen at the nest on February 3, 1946; careful examination of the nest began when the one egg hatched on March 7 and continued until April 25, shortly before the young owl left the nest.

One large cottonwood tree, used by the parent-owls as a landing place whenever they were forced from the nest, was situated approximately 110 feet to the north and a five-story building was located 80 feet farther to the north. Numerous smaller trees line the street to the east and there are some on the lawns around the Museum. Also, there are about two acres of trees 225 feet west of the nest-site where the parent-owls took refuge when forced from the cottonwood tree.

The nest, if it can be called a nest, was no more than a few bare branches of the Virginia creeper, which covers the side of the building, together with some excrement which the owls tended to push to the periphery of the nest. For most of the time the three eggs in 1945 lay directly on the metal which covered the ledge, because there was no definite floor to the nest. The single egg in 1946 lay on the cement shelf between the pillars and the wall of the building. This laxity in nest building by Great Horned Owls apparently is not uncommon (see Bent, 1938:300).

PERIOD OF INCUBATION

Incubation of the eggs probably began, in 1945, on February 5, the day the first egg was laid. It has usually been assumed that, in birds of prey, incubation begins when the first egg is laid. The last of the three eggs was laid February 7. In 1946, the single egg was being incubated on February 4. Since another egg had been laid two or three days before this a broken egg was found beneath the nest and there were remnants of the egg in the nest incubation may have started as early as February 1 or February 2. In comparing these dates of initial incubation with other recorded dates of nesting, only those from places at, or near, the latitude of Lawrence, Kansas, in the central United States, should be expected to be approximately the same since the times of egg-laying and incubation are progressively later in the year as approach is made toward the polar region. Baumgartner (1938:279) has previously pointed this out.

The incubation period for the Great Horned Owl in the central United States has usually been regarded as 28 to 29 days. In the nest under observation in 1945, two eggs hatched on March 12 and are assumed to be the first two eggs laid, with an incubation period for each of 35 and 34 days, respectively, and the third egg hatched on March 14 with an incubation period of 35 days. In 1946, the single egg hatched on the 33rd day, assuming that incubation began on February 2, for the egg hatched March 7. In the period of egg-laying and also in incubation, the parent bird in 1945 was frequently disturbed by persons who

peered at it through the window. Curious observers handled the eggs at least once and vigorous pounding by carpenters in the room adjacent to the nest frequently flushed the adult bird but did not cause desertion of the nest. It may be that such disturbances prolonged the incubation period. However, in 1946, the brooding birds were undisturbed, yet the incubation period was nearly as long. If an observer near the nest exposed himself in the daytime to the incubating bird, the adult flew, but exposure at 50 feet or more from the nest only caused the incubating bird to remain alert on the nest. When flushed, the parent usually returned to the nest within 15 minutes or less after the observer withdrew. On the thirty-second and thirty-third days of incubation in 1945, the crew of carpenters demolished partitions within the building on which the owl was nesting, and within 15 feet of the nest itself. At first the adult would fly from the nest at each outburst of hammering and, at one time, remained away from the nest for more than two hours. After a few hours of intermittent hammering, however, the parent bird remained on the nest despite all the noise produced. These observations bear out, rather than refute, Baumgartner's statement (1938:281) that "the horned owl incubates very closely," for a strong stimulus was necessary to keep the owl from covering the eggs.

The egg hatched on March 14, 1945, and approximately two days later than the other two, is judged to be the one laid last. This owl, III, was always 5 to 21 per cent lighter in weight than the older birds when weights for corresponding ages were compared. Whether this difference was the result of a lack of food because of dominance of the two older birds, or because of a sexual difference, we do not know. The owl that hatched in 1946 was likewise markedly lighter than the first two birds hatched in 1945 (figure 1). A series of adults from Meade County, southwestern Kansas, shows a pronounced secondary sexual difference in weight. In this sample the mean weight of 17 males, 1,208 grams, was 21 per cent less than that of 25 females, 1,531 grams.

GROWTH OF JUVENILES

The principal measurement of growth taken by us was the weight of the owls. In 1945 each of the three owls was weighed daily, with two or three exceptions when a 48-hour period was interposed between weighings. The young were removed from the nest to a nearby balance, weighed, and examined. The owl last hatched (owl III) was weighed on the first day of life and on most subsequent days. The other two owls (designated as owls I and II) were first weighed when they were between 53 and 60 hours old. On some days the birds

were weighed twice, once in the morning and once in the late afternoon; on most days, they were weighed only in the late afternoon. The owl hatched in 1946 was weighed when seven days old and at irregular, but usually two day, intervals thereafter. It was weighed always slightly before midday.

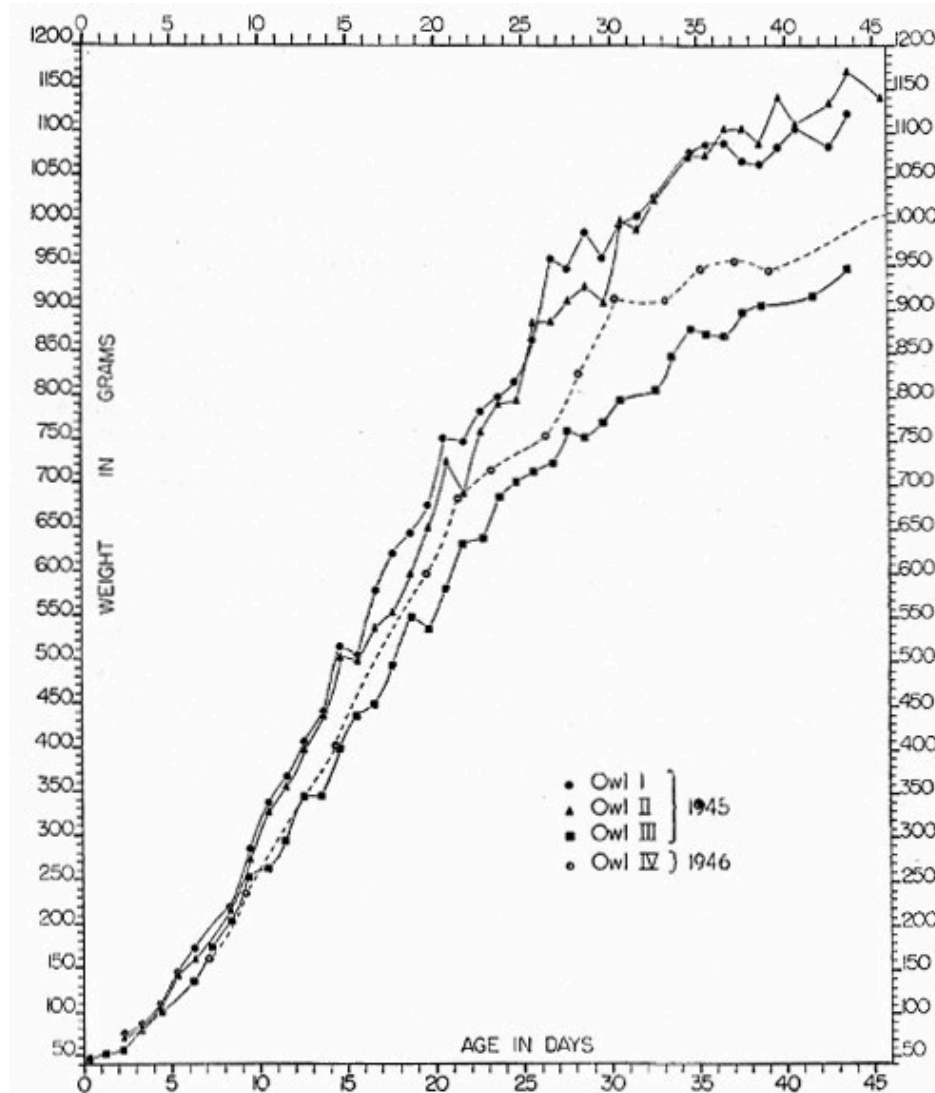


Fig. 1. Growth of four Great Horned Owls as shown by changes in weight from near the time of hatching until the time of leaving the nest.

The growth of the four owls is well shown by the changes in weight recorded in figure 1. For the period during which the young owls remained at the nest, growth can be divided into two phases: (1) a rapid increase in weight during the first 3-1/2 or 4 weeks while the parent birds are supplying the young with ample food; and (2) a subsequent period of slower growth, marked by fluctuations (actual losses as well as gains) in weight resulting from the failure of the parent

birds to provide an ample supply of food. If there is an initial period of about one week in postnatal development in which there is a rather slow gain in weight, as suggested by Sumner (1933:284), it was poorly marked in this instance. Owl IV remained at the nest until the 50th day of age, and on the 47th and 49th days (not shown on chart, fig. 1) weighed 1,011.4 grams and 971.4 grams, respectively. By this age, the growth curve had definitely flattened out. The fact that owl IV was consistently heavier than owl III might be accounted for, in part, by the fact that owl IV was always weighed in the morning when it was gorged with food. However, Riddle, Charles, and Cauthen (1932) have pointed out that when there were two or more pigeons in a nest, each grew less rapidly than if there was only one present.

Within about 12 hours after hatching, the smallest of the three owlets (III) weighed 48.7 grams. During the first four weeks of postnatal growth, each owl gained in weight, daily, an average of $33\frac{1}{3}$ grams or an increase of 11.1 per cent. Owl I gained an average of 36.1 grams each day, or a daily increase of 10.7 per cent; owl II, 37.8 grams, or 11.2 per cent; and owl III, 26.1 grams, or 11.4 per cent. From the beginning of the fifth week until the time the young left the nest, the three owls gained on the average only 12.7 grams or approximately 1.6 per cent in weight daily. Individually, the daily mean increases were as follows: I, 9.6 grams or 0.93 per cent; II, 12.7 grams or 1.86 per cent; III, 15.8 grams or 1.97 per cent. Prior to the twenty-sixth to twenty-eighth day of age, there seldom was any loss in weight from day to day, whereas after this period, approximately one weight in four was less than on the previous day. These data support the earlier statement that during the first $3\frac{1}{2}$ or 4 weeks, there is a relatively uniform and rapid increase in weight whereas after this period weight fluctuates.

Growth as measured by changes in weight in these young Great Horned Owls parallels growth in some other young birds. For example, nestling Red-tailed Hawks, as reported upon by Fitch, Swenson, and Tillotson (1946:215), increase in weight rapidly for about the first three weeks and then more gradually. Sumner's (1929b) graphs indicate the same pattern of growth in the Barn and Great Horned owls and Red-tailed and Cooper hawks. Pigeons, judging from the growth curves for bodily weight as given by Riddle, Charles, and Cauthen (1932), increase in weight rapidly until somewhere between the twenty-fourth and thirty-second day of postnatal development. However, in the Golden Eagle, the early part of postnatal development is not one of rapid growth, judging from Sumner's diagram (1929a:164), but after the fourth week there is a rapid increase in weight. Graphs that Sumner (1929b) gives for Sparrow Hawks,

Long-eared Owls, and Screech Owls, indicate that in these instances also the increase in weight during the first few days of postnatal development was not so rapid as it was after the end of the first week. Stoner (1935) indicates that in the young Barn Swallow, increase in weight was most rapid between the fourth and tenth days, with the young remaining at the nest until the twentieth day. Much the same pattern of weight increase was noted by Stoner (1945) in the Cliff Swallow. Huggins' (1940:228) sigmoid curve for increase in weight in the House Wren indicates that the period of rapid growth in this species does not begin until the second day. Sumner (1934:284) cites other studies which he believes, for altricial birds, indicate three periods of growth, an initial period of rather slow gain, a period of maximum increase in weight, and a final period of fluctuations. As previously indicated, for the Great Horned Owls under observation, and in some other species as indicated by published growth curves, the initial period of slow gain is lacking.

The period of a decelerated rate of growth in the young Great Horned Owls is correlated with the occasional lack of food. The parent birds, during this latter period, remain off the nest more of the time during the day, and their failure to provide the young with food may represent an attempt to force the young to become proficient in flight or to force them away from the nest site, which amounts to the same thing. When only slightly more than a month old, the young began to test their wings, springing into the air, and, in general, becoming more active and alert. Sumner (1929b:110) has suggested some other possible reasons for the period of decelerated rate of growth.

Although there was a daily increase in weight in the early stages of growth, there was a decided fluctuation in any twenty-four hour period. On any given day, the young always were heavier in the morning than in the afternoon (see figure 2); presumably they were gorged with food early in the morning.

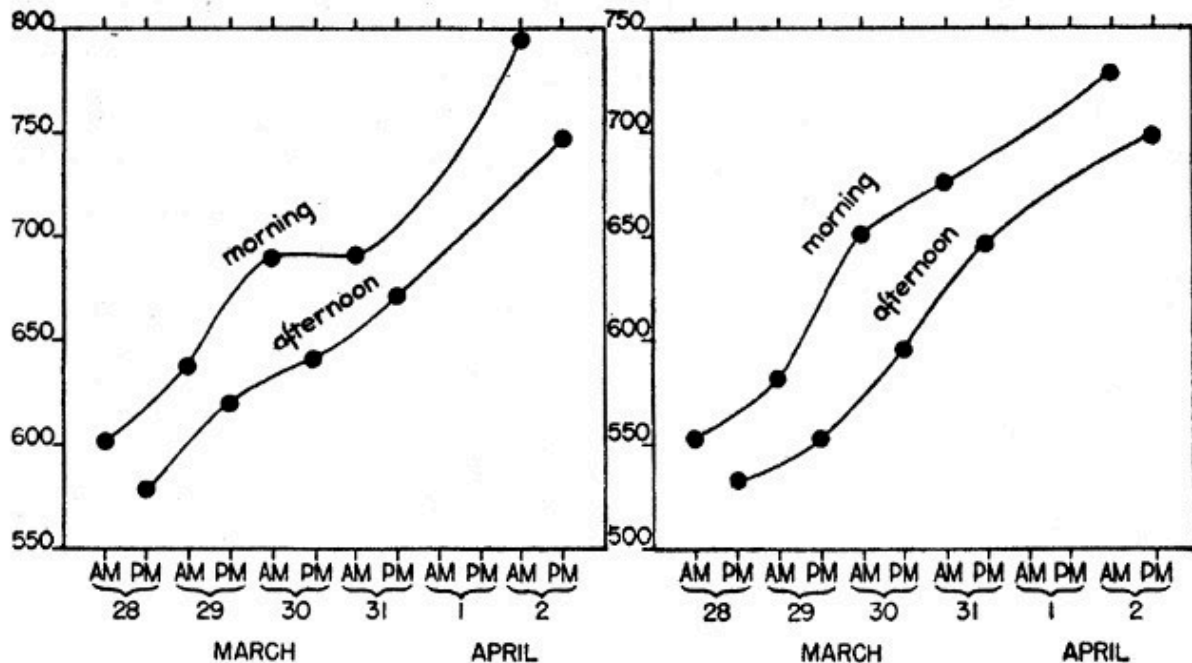


Fig. 2. Morning and afternoon weights of two Great Horned Owls. Note that in the morning the owls weigh more than in the afternoon.

When the young left the nest, they were approximately three-fourths grown. When owl I on the 44th day and owl II on the 45th day left the nest, they weighed 1,120 and 1,139 grams, respectively, or 73 and 74 per cent, respectively, of the average weight of 25 females (1,530.9 grams). Owl III weighed 943.3 grams on the 43rd day and owl IV weighed 971.4 grams on the 49th day, or 78 and 80 per cent, respectively, of the average weight of 17 mature males (1,207.7 grams). Owl I left the nest 18 hours before owl II did. Owl III attempted to leave when 43 days old, but for it coördinated flight was impossible and the bird landed on the lawn after a 150-foot glide. When attempting to take owl IV from the nest on the 49th day, it sprang into the air and by gliding, aided by an occasional flap, sailed more than 300 feet before alighting on the ground. After we returned the owl to the nest, it immediately sailed forth for even a longer distance. When attempt was made to pick up owls III and IV after they had flown down to the ground, they rolled over on their backs and used both claws and beaks defensively. Such a reaction never was noted at the nest; there our hands were inspected, and sometimes bitten by the owls as possible sources of food, but the claws were rarely used offensively or defensively.

Slightly elevated remiges and rectrices, still in the sheath, were visible on the ninth day. Some remiges first ruptured the feather sheath on the 14th day; nearly

all of the primaries ruptured the sheaths by the 19th day and the secondaries by the 20th day. The amount of eruption from the sheath for primaries, secondaries, and rectrices, as given in table 1, was determined by measuring the one feather of, say, the secondaries, judged to be near the mean in degree of eruption. The feathers of the wing at 21 and 47 days of age are shown in figure 5. On the eighth day, or slightly before, the white nestling down of the newly hatched bird was replaced by a downy immature plumage, which was more yellowish than the preceding plumage. The development of the plumage in the birds under observation was much the same as that recorded by Sumner (1933) in *Bubo virginianus pacificus* Cassin.

TABLE 1.Changes with age in certain parts of a young Great Horned Owl hatched in 1946.

(Measurements are in millimeters)

Age in days	19	21	26	33	37	39	49
Length of erupted portion of "average" primary	6.0	10.0	26.0	93.0	87.5	99.2	
Length of erupted portion of "average" secondary		5.0	25.0	60.0	78.0	95.0	
Length of erupted portion of "average" rectrix	17.0		16.0	28.0			78.0
Length of exposed culmen without cere		19.6		22.5		24.0	23.7
Length of total culmen		30.4		36.0		38.5	40.0
Length of femur	69.0			87.5		89.5	

Traces of the egg-tooth were retained until the ninth day in two owls and until the 11th day in another. In owl IV, the egg-tooth was lost sometime between the 9th and 14th day. Changes in the length of the culmen are indicated in Table 1. The length of total culmen of owl IV when 47 days old is slightly greater than the average for three adult male owls from Lawrence, Kansas (40.0 mm. as contrasted with 39.2 mm. in the adults). The length of exposed culmen, without cere, in the same bird when 47 days old is less than the average of this measurement in the three adults from Lawrence (23.7 mm. as contrasted with 26.5 mm. in the adults). The femur was measured on three occasions as accurately as possible through the skin and flesh. The precise boundaries of the femur could not be determined and the thickness of the skin and certain muscle

is included. These measurements are not given to indicate actual length of the femur, but to indicate the relative changes in length of this bone.

Remnants of the yolk stalk were clearly evident at seven days of age (see fig. 5) in the owl hatched in 1946 and were still present when the owl was last examined (49 days of age) just before the young left the nest. The scablike remnant was not noted in the three young hatched in 1945, but close inspection was not made to see if it was present.

The instinctive reactions of young horned owls shortly after hatching have been fully described by Sumner (1934). By the third day our owls could raise their heads, but when a bird was undisturbed its head lay on the nest floor and the wings were slightly spread. The eyes of owls I and II opened at about 7-1/2 days, those of owl III on the 6th day, and those of owl IV sometime between the 7th and 9th days. After the eyes opened, the head was held erect more of the time. The young responded with "cries" when disturbed by handling, when stimulated by certain movements of the parent, or by movements of our hands near their heads, which suggested to them the possibility of being fed. Cries were evident but weak in the unhatched, pipped, egg, but soon after hatching increased in intensity, and beginning at six days of age were replaced, in part, by the characteristic "bill-snapping" of more mature birds. These cries of the young may serve, among other things, for recognition, inasmuch as they were given when the parent was inspecting the young. When the parent returned to the nest and covered the young, after having been flushed, it sometimes uttered a special note, "hut, hut, hut," much like the "cluck" note of the hen of the domestic chicken. The young responded to these notes with faint cries, in contrast to the loud cries signifying alarm and possibly hunger, which they gave when the parents were absent from the nest.

The first definite evidence that the young were attempting to feed themselves was obtained when they were 23 days old. Frequently thereafter, fresh blood was found on their beaks and claws, but as late as the 34th day a parent was seen feeding them. That day, after being flushed, a parent returned to the nest at 7 p. m., and began tearing away parts of a cottontail which had previously been brought to the nest. Bones in the hind leg of the rabbit broke readily under pressure of the parent's bill, and the three young crowded in close, opening their bills widely and placing them around that of the parent. Of cottontails, the only parts consistently uneaten were the upper cheek teeth and the supporting maxillae and connecting palatal bridge.

FOOD BROUGHT TO THE NEST

In the 45 days that the young remained at the nest site in 1945, ninety-one individuals of 16 different species of animals were brought by the parent owls (table 2). Probably a few smaller animals, of which we saw no traces, were caught and eaten at night. In 1946, two additional kinds of birds were brought to the nest: 1 Baldpate (*Mareca americana*) and 1 Pied-billed Grebe (*Podilymbus podiceps*). The large number of Rock Doves in the list can be explained by their abundance on the buildings on the University campus, including the Museum building where some were nesting as close as 100 feet to the owl nest. When the owls were less than a week old, only small birds and mammals were brought (young Rock Doves, Robins, Starlings, Grasshopper Sparrow, meadow mouse, and Norway rat). The first rabbit was brought when the owls were eight days old.

TABLE 2. Number of food items brought to the nest by the Great Horned Owls in 1945

Birds	
Rock Dove (<i>Columba livia</i>)	32
Robin (<i>Turdus migratorius</i>)	6
Starling (<i>Sturnus vulgaris</i>)	4
Mourning Dove (<i>Zenaidura macroura</i>)	10
Meadowlark (<i>Sturnella sp.</i>)	3
Red-wing (<i>Agelaius phoeniceus</i>)	1
Bronzed Grackle (<i>Quiscalus versicolor</i>)	1
Mockingbird (<i>Mimus polyglottos</i>)	1
Brown Thrasher (<i>Toxostoma rufum</i>)	1
Grasshopper Sparrow (<i>Ammodramus savannarum</i>)	1
Coot (<i>Fulica americana</i>)	3
Sora (<i>Porzana carolina</i>)	1
Blue-winged Teal (<i>Anas discors</i>)	1
Mammals	
<i>Sylvilagus floridanus</i>	19
<i>Rattus norvegicus</i>	6
<i>Microtus ochrogaster</i>	1

After the 28th day, only 18 food items, or slightly less than 20 per cent of the total number, were brought to the nest. These last 18 food items brought after the owls were 4 weeks old were no larger or bulkier than those brought in the previous 20 days. The beginning of this period of reduced amount of food corresponds to the beginning of the second phase of growth characterized by marked fluctuations in weight.

Fox squirrels (*Sciurus niger*) are abundant on the University campus, yet there were no remains of this mammal at the nest. This may be explained by the fact that fox squirrels are principally diurnal and Great Horned Owls feed principally at night. Yet in early February, 1946, when the owls were preparing the nest, they frequently flew on and off the nest in the early twilight of evening while one or two fox squirrels fed in the periphery of trees not more than 25 feet away. Yet the owls flew off to the west and left this source of food unmolested.

Whether both owls regularly attended the young we do not know, for the adults were not distinctively marked. On March 17, 1945, when weighing the young, one parent bird started to return to the nest but was frightened away by the observer who at the same time noted the other parent perched in an adjacent tree. This was the first time two adults were seen at the same time near the nest. In 1946, two adult owls (presumably both were parents) were within sight at one time when the young owl first sailed forth and landed in a wooded area some 100 yards away.

SUMMARY

Great Horned Owls (*Bubo virginianus virginianus*) have employed as nest sites the protruding shelves of the stone wall of the Museum of Natural History at the University of Kansas for several years. In 1945, daily observations were made on one such nest and its three young, and in 1946 irregular observations were made on another such nest and the one young owl. The incubation time for the three owls, hatched in 1945, was 35 days for two of the young and 34 days for the third; for the one owl hatched in 1946, the incubation time was at least 33 days. Two owls were consistently smaller; when these smaller two left the nest they were, respectively, 21 and 17 per cent lighter than the other two. The smaller two were judged to be males because adult males in Kansas average smaller by 21 per cent than adult females.

Growth of the nestling young is divisible into (1) a period of rapid increase in weight during the first 25 to 28 days, and (2) a subsequent period marked by

gains and losses in weight. The fluctuations in this latter period are correlated with a reduction in food brought to the nest by the parent birds and with the development of habits of flight. This second period may be considered to be a period of "weaning." By the forty-fifth day, the young owls are able to fly short distances and thus are able to leave the site of the nest permanently. At this time they are about three-fourths grown.

Ninety-one individuals of 16 species of birds and mammals made up the food items brought to the nest in 1945. Two factors seem to be concerned in the acquisition of prey: (1) its availability and (2) appropriate size of the prey.

LITERATURE CITED

BAUMGARTNER, F. M.

1938. Courtship and nesting of the Great Horned Owls. *Wilson Bull.*, 50:274-285.

BENT, A. C.

1938. Life histories of North American birds of prey (Part 2), Orders Falconiformes and Strigiformes. U. S. Nat. Mus., Bull. 170, viii + 482 pp.

FITCH, H. S., SWENSON, F., and TILLOTSON, D. F.

1946. Behavior and food habits of the Red-tailed Hawk. *Condor*, 48:205-237.

HUGGINS, S. E.

1940. Relative growth in the House Wren. *Growth*, 4:225-236.

RIDDLE, O., CHARLES, D. R., and CAUTHEN, G. E.

1932. Relative growth rates in large and small races of pigeons. *Proc. Soc. Exp. Biol. and Med.*, 29:1216-1220.

STONER, D.

1935. Temperature and growth studies on the Barn Swallow. *Auk*, 52:400-407.

1945. Temperature and growth studies of the Northern Cliff Swallow. *Auk*, 62:207-216.

SUMNER, E. L., JR.

1929a. Notes on the growth and behavior of young Golden Eagles. *Auk*, 46:161-169.

1929b. Comparative studies in the growth of young raptors. *Condor*, 31:85-111.

1933. The growth of some young raptorial birds. *Univ. California Publ. Zoöl.*, 40:277-307.

1934. The behavior of some young raptorial birds. Univ. California Publ. Zoöl., 40:331-361.

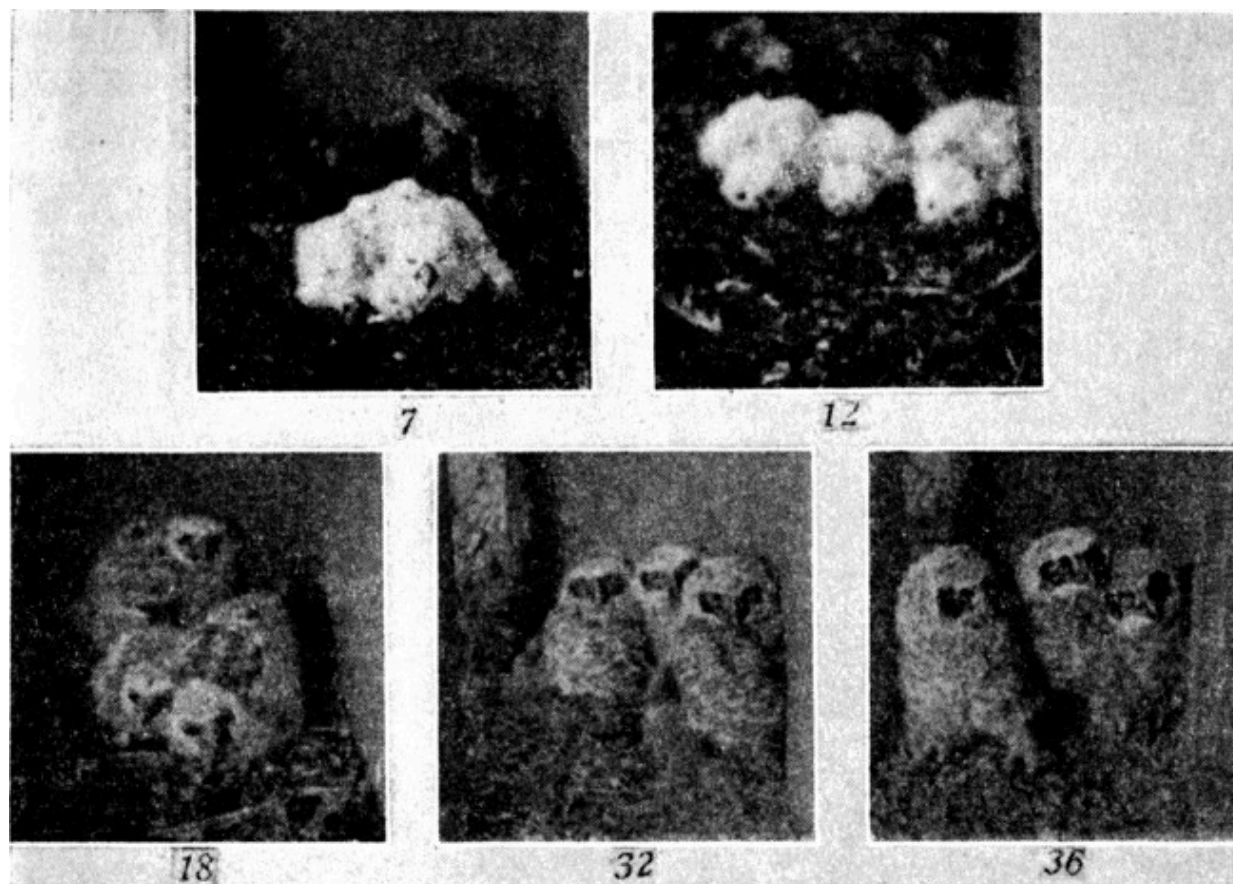


Fig. 3. Young Great Horned Owls in nest. Two owls are 7, 12, 18, 32, and 36 days of age, respectively; the third owl is about 2 days younger in each instance.

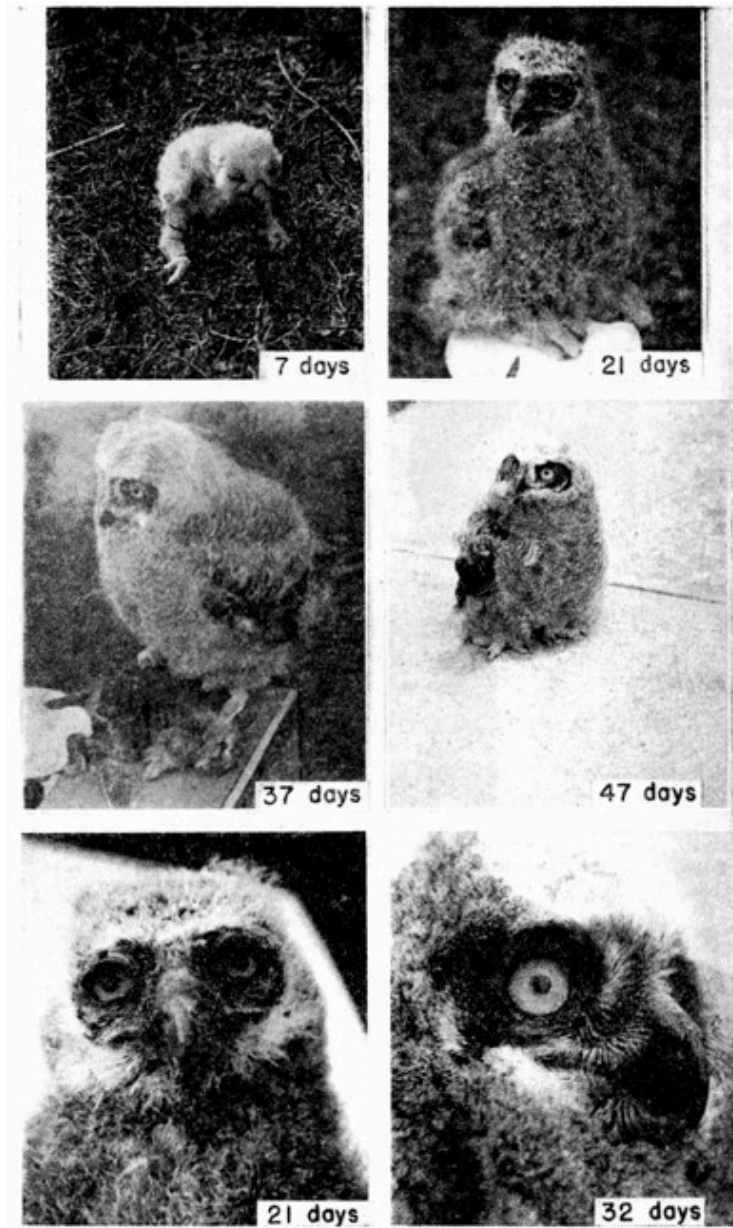


Fig. 4. Young Great Horned Owl hatched in 1946. The two lower pictures show the developing facial mask. Photographs by João Moojen.

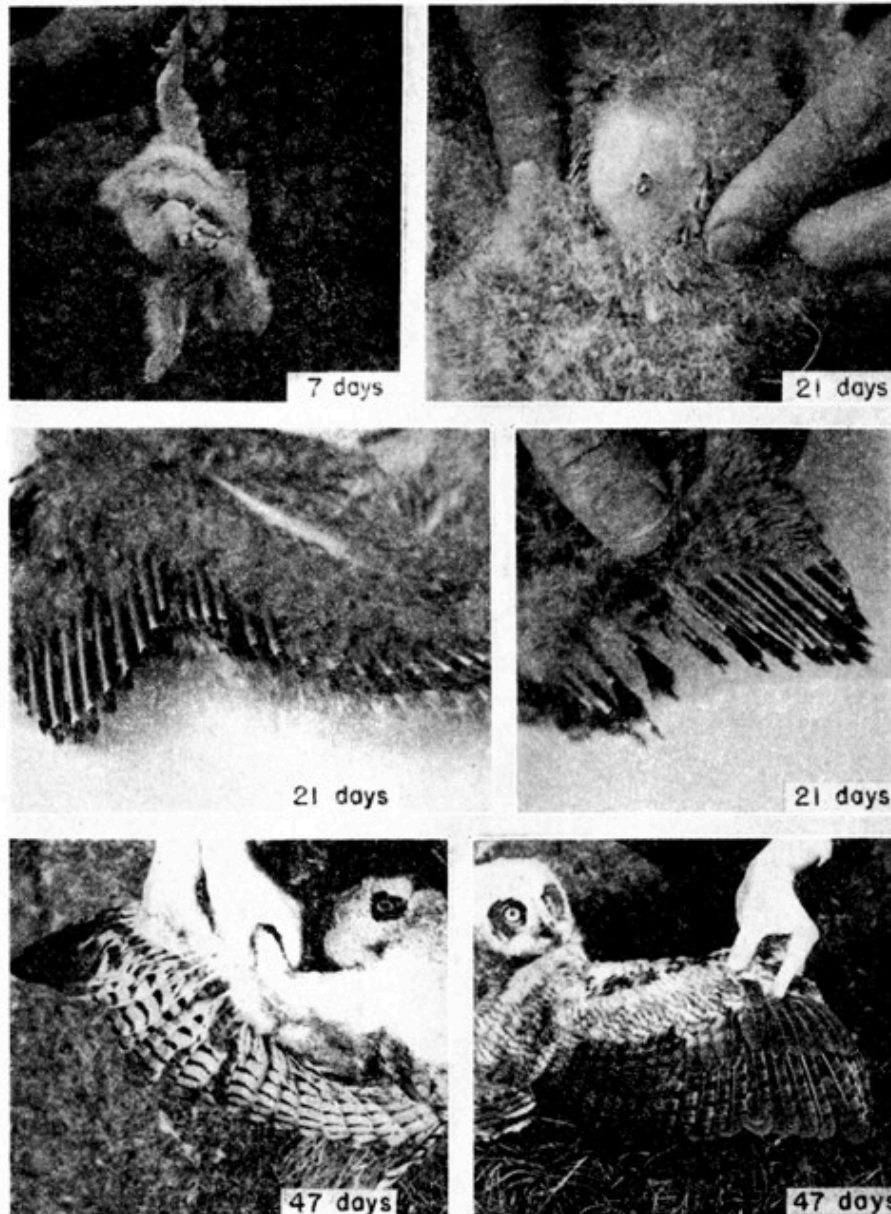


Fig. 5. Young Great Horned Owl hatched in 1946. Upper row: Ventral views showing scar of yolk sac and ventral side of wing. Middle row: Ventral (left) and dorsal view of wing at 21 days. Bottom row: Ventral (left) and dorsal view of wing at 47 days. Photographs by João Moojen.

21-6958

*** END OF THE PROJECT GUTENBERG EBOOK THE POSTNATAL
DEVELOPMENT OF TWO BROODS OF GREAT HORNED OWLS
(BUBO VIRGINIANUS) ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.